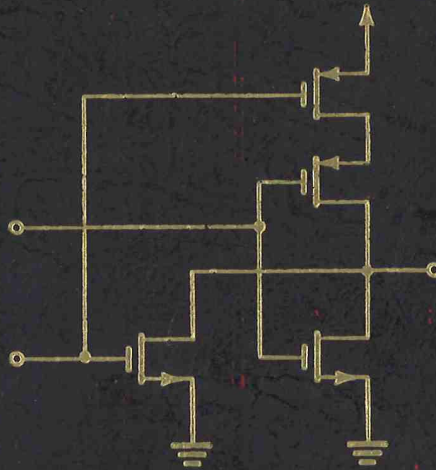


EXHIBIT 31

Adel S. Sedra
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Sedra
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MICRO-
ELECTRONIC CIRCUITS

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MICROELECTRONIC CIRCUITS

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METAL-OXIDE- 8 SEMICONDUCTOR FIELD-EFFECT TRANSISTORS (MOSFETs)

In this chapter we study a very popular class of field-effect transistor, the metal-oxide-semiconductor FET, or, simply, MOSFET. As will be seen, digital logic and memory functions can be implemented with circuits that exclusively use MOSFETs (that is, no resistors or diodes are needed). For this reason, because MOS devices can be made quite small (that is, occupying a small silicon area on the IC chip), and because their manufacturing process is relatively simple (as compared to that of bipolar transistors; see Appendix A), most very-large-scale-integrated (VLSI) circuits are currently made using MOS technology. Examples include microprocessor and memory packages. More recently (1977), MOS technology has been shown to be applicable in the design of analog integrated circuits. For instance, precision filter networks (Chapter 14) have been manufactured in integrated-circuit form using MOS op amps, MOS capacitors, and MOS analog switches. Nevertheless the bipolar-junction transistor (Chapter 9) remains the most suitable device for the design of precise linear circuits (such as op amps).

MOS transistors are either of the *p*-channel or *n*-channel type. *p*-channel devices (PMOS) were initially quite popular in digital applications, but more recently the NMOS (*n*-channel) technology has been perfected. The latter technology results in faster devices which occupy smaller silicon area. Therefore NMOS technology has become dominant in VLSI applications.

Another popular MOS technology is called

Introduction

complementary-symmetry MOS (COS MOS or CMOS). As the name implies, CMOS circuits use both n -channel and p -channel devices on the same IC chip. CMOS logic circuits are currently quite popular and are sometimes referred to as the “ideal” logic family.

MOS transistors have very high input impedance and consume little static power. This makes them quite useful in the design of micropower circuits, both digital and linear. Needless to say, the MOS transistor is also very useful in the design of amplifiers with extremely high input impedance. MOS transistors can also be used as analog switches. This together with the ability to manufacture capacitors whose ratios are very accurate using MOS technology has motivated its application in the design of analog signal-processing circuits (see Chapter 14).

Before proceeding the reader is encouraged to read Appendix A in order to gain some familiarity with MOS technology.

8.1 THE DEPLETION-TYPE MOSFET

Physical Structure

Unlike JFETs, which are all of the depletion type, MOSFETs can be either depletion or enhancement type (the exact meaning of these words will become clearer shortly). The operation of the depletion-type MOSFET is very similar to that of the JFET. Let us start by considering the depletion-type MOSFET shown in Fig. 8.1. As shown, the

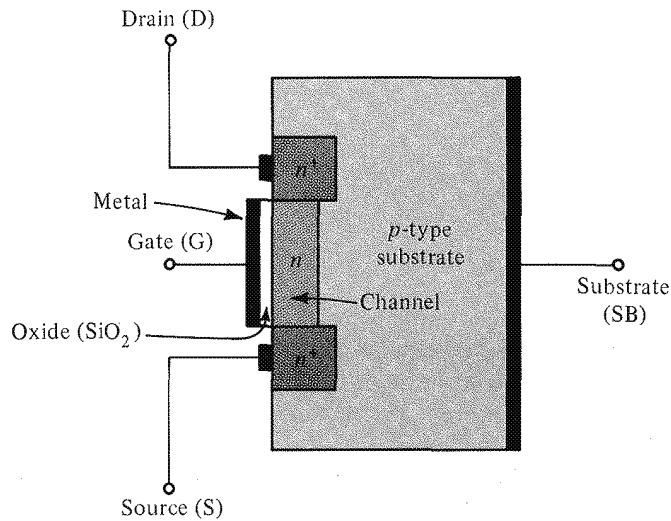


Fig. 8.1 Physical structure of n -channel MOSFET of the depletion type.

n -channel device is formed on a p -type silicon substrate. The two heavily doped n^+ wells form low-resistance connections between the ends of the n channel and the metal contacts of the source (S) and the drain (D). A thin oxide layer is grown on the surface of the channel, and metal (aluminum) is deposited on it to form the gate (G). It should be clear that the name MOS is derived from the structure of the device.

While the JFET is controlled by the gate-channel pn junction, no such junction exists in the MOSFET. Here the oxide layer acts as an insulator that causes the gate current to be negligibly small (10^{-12} to 10^{-15} A). This gives the MOS transistor its extremely high input resistance under all conditions.

Circuit Symbol

Figure 8.2 shows the circuit symbol of the n -channel depletion-type MOSFET. Note that the space between the line representing the gate electrode and the line representing

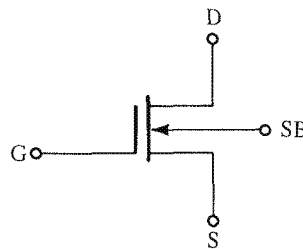


Fig. 8.2 Circuit symbol for the n -channel depletion-type MOSFET.

the channel represents the insulating oxide layer. As in the JFET symbol the gate terminal is drawn closer to the source than to the drain. The arrow on the substrate line points in the forward direction of the substrate-to-channel pn junction and hence indicates the type (or polarity) of the device (n - or p -channel device).

In most applications the substrate is electrically connected to the source. Since the drain voltage will be positive with respect to the source, the substrate-to-channel junction will always be reverse-biased and hence the substrate current will be almost zero. In the following we will normally ignore the role of the substrate in our description of MOSFET operation.

A simplified symbol for the n -channel MOSFET is shown in Fig. 8.3. Note that the polarity of the device is indicated by the direction of the arrowhead on the source line; the arrowhead points in the normal direction of current flow in the source lead.

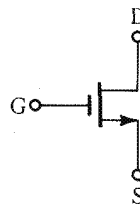


Fig. 8.3 Simplified circuit symbol for n -channel MOSFET.

Physical Operation

As mentioned before, the depletion-type MOSFET operates much like the JFET. The fundamental difference is that in the JFET the channel is depleted by reverse-biasing the gate-to-channel junction. In the MOSFET control of the channel width is effected by an electric field created by the voltage at the gate.

Consider first the triode—or voltage-controlled resistance—region of operation. Let v_{DS} be small while v_{GS} is either negative or positive (note the difference between this case and the JFET case, where v_{GS} has to be negative or at most 0.5 V positive). If v_{GS} is negative, some of the electrons in the n -channel area will be repelled from the channel, and thus a depletion region will be created below the oxide layer. The result will be a narrower channel, as shown in Fig. 8.4. From Fig. 8.4 we also note that the

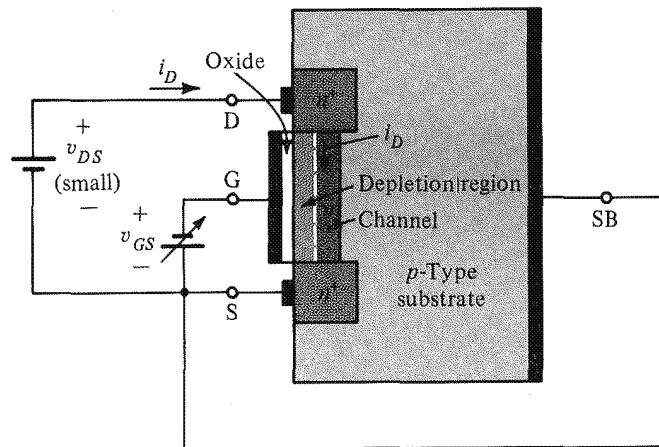


Fig. 8.4 Physical operation of the depletion-type MOSFET for small v_{DS} .

channel has a uniform width because $v_{GS} \approx v_{GD}$. The narrower channel will have a higher resistance r_{DS} , giving rise to the set of straight-line characteristics shown in Fig. 8.5.

As v_{GS} is made more negative, a point will be reached at which the channel is completely depleted of charge carriers. The value of v_{GS} at which this happens is called the pinch-off voltage V_p ,

$$V_p = v_{GS} |_{i_D=0, v_{DS}=\text{small}}$$

Obviously for an n -channel device V_p is a negative number.

Unlike the JFET case, v_{GS} can be made positive, with the result that the channel becomes wider and its resistance decreases, as illustrated in Fig. 8.5.

Next consider the operation as v_{DS} is increased. If v_{GS} is kept constant while v_{DS} is increased, the depletion region will have a constant width at the source end, but it will be progressively wider as we move toward the drain. Thus the channel will have a tapered shape, and its width will be narrowest at the drain end. It follows that as v_{DS} is increased the channel resistance increases, giving rise to a nonlinear (parabolic) i_D -

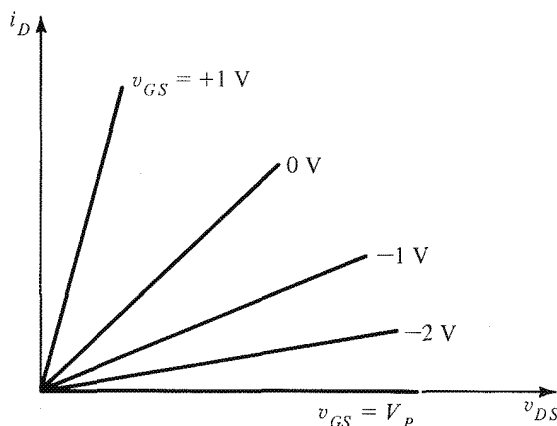


Fig. 8.5 The i_D - v_{DS} characteristics of the depletion-type MOSFET for small v_{DS} .

v_{DS} curve. Finally, if v_{DS} is increased to the value at which $v_{GD} = V_p$, the channel will be pinched off at the drain end. Any further increase in v_{DS} does not alter the channel shape, and hence the current in the channel remains constant at the value reached for $v_{GD} = V_p$ (or, correspondingly, $v_{DS} = -V_p + v_{GS}$). The result is the i_D - v_{DS} characteristic curves shown in Fig. 8.6.

Static Characteristics

The static characteristics of the n -channel depletion-type MOSFET take the form shown in Fig. 8.6. These characteristics are identical to those of the JFET except that in the MOSFET positive values for v_{GS} are allowed. A positive v_{GS} “enhances” the channel by attracting more electrons into it, thus increasing its width and reducing its resistance. A depletion-type MOSFET can therefore be applied in the *enhancement mode*. Figure 8.7 illustrates this point further by showing the i_D - v_{GS} characteristic curve for a device operating in the pinch-off region ($v_{DG} > |V_p|$).

The equations describing the characteristic curves of Figs. 8.6 and 8.7 are identical to those of the JFET (Chapter 7) and will not be repeated. Thus with the exceptions noted above the depletion-type MOSFET behaves much like the JFET and can be applied using circuit techniques similar to those discussed for the JFET.

The characteristic curves in Fig. 8.6 assume that in pinch-off the device acts as an ideal constant-current source. Real MOSFETs, however, have a finite output resistance, with the result that their pinch-off i_D - v_{DS} characteristics have finite nonzero slopes. Furthermore, the output resistance will decrease as the current level in the device is increased.

EXERCISE

- 8.1 Consider an NMOS transistor with $V_p = -2$ V and $I_{DSS} = 8$ mA. Find the minimum v_{DS} required for the device to operate in pinch-off when $v_{GS} = +1$ V. What is the corresponding value of i_D ?

Ans. 3 V; 18 mA

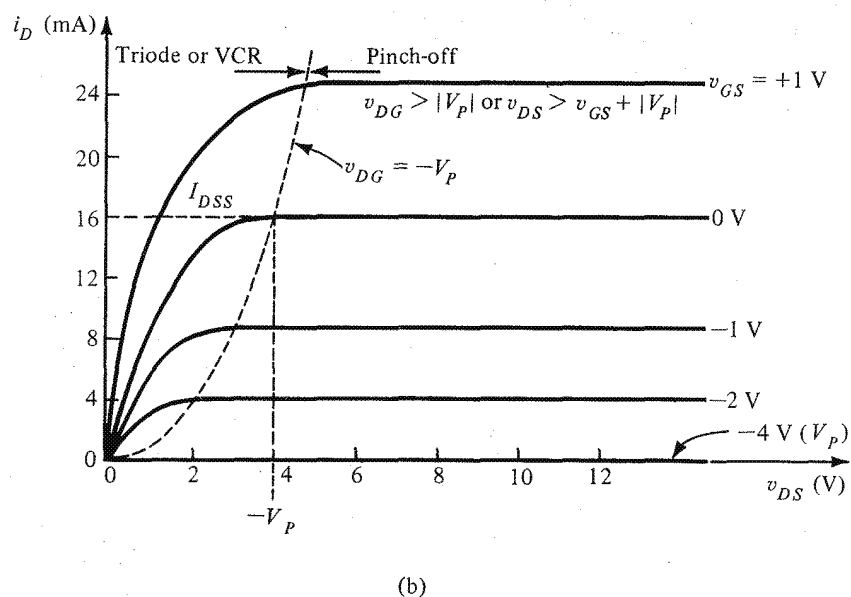
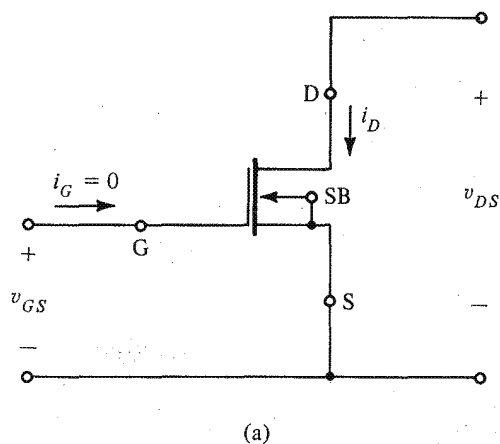


Fig. 8.6 The i_D - v_{DS} characteristics for a depletion-type MOSFET whose $V_P = -4$ V and $I_{DSS} = 16$ mA.

8.2 THE ENHANCEMENT-TYPE MOSFET

Structure and Physical Operation

Figure 8.8 shows an enhancement-type MOSFET. As seen, the structure looks quite similar to that of the depletion device in Fig. 8.1, with one major exception: there is no channel. It follows that if the gate is left floating, or if $v_{GS} = 0$, the path from drain to

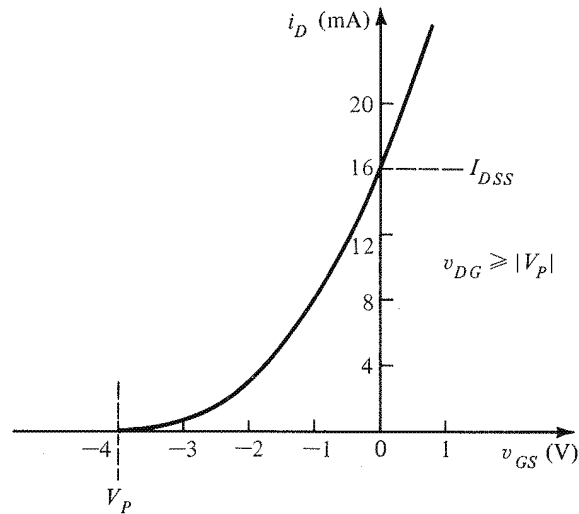


Fig. 8.7 The i_D - v_{GS} characteristic of an n-channel depletion-type MOSFET operating in pinch-off ($V_p = -4$ V, $I_{DSS} = 16$ mA).

source includes two series diodes back-to-back, which means that no current can flow. To cause current to flow from drain to source we first have to create an n channel. This can be done by applying a positive voltage v_{GS} . Such a positive voltage at the gate will attract electrons from the substrate and cause them to accumulate at the surface beneath the oxide layer. To attract sufficient number of electrons to form an n channel, the voltage v_{GS} has to become equal to or greater than a threshold value V_T . In other words, no appreciable current i_{DS} will flow until v_{GS} equals V_T .

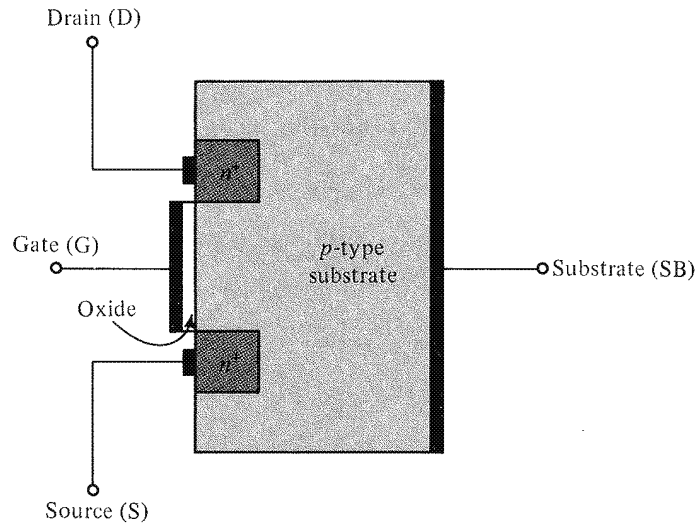


Fig. 8.8 Physical structure of n-channel enhancement-type MOSFET.

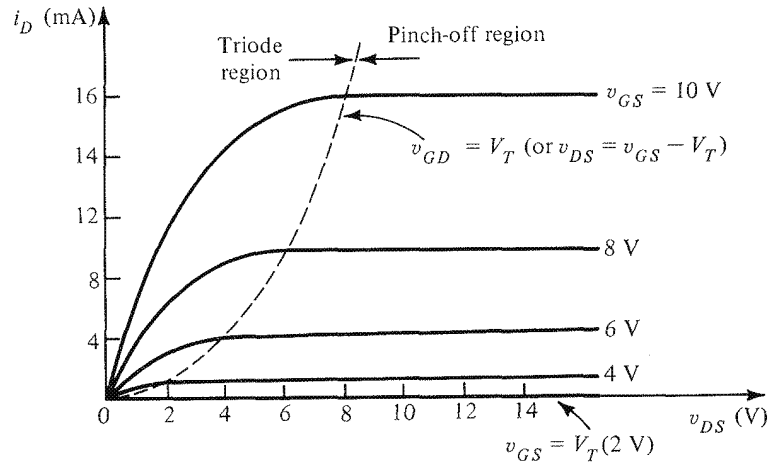


Fig. 8.9 Ideal static i_D - v_{DS} characteristics for an n -channel enhancement-type MOSFET whose $V_T = 2$ V and $\beta = 0.5$ mA/V².

Consider now the case where v_{DS} is small, and let v_{GS} be increased above V_T . Increasing v_{GS} will cause the created channel to widen (become *enhanced*), and thus its resistance will decrease. This process gives rise to the voltage-controlled-resistance (VCR) or triode region of operation, indicated in Fig. 8.9.

Next consider what happens if we keep v_{GS} constant—say, $v_{GS} = V_T + V$, where V is a positive voltage—and increase v_{DS} . The channel will remain at a constant width at the source end, since v_{GS} is constant. However, since increasing v_{DS} means that v_{DG} increases or, equivalently, v_{GD} decreases, the channel will become narrower at the drain end. In fact, the channel will have a tapered shape, being widest at the source end and narrowest at the drain end. It follows that increasing v_{DS} will cause the channel resistance to increase, giving rise to a nonlinear i_D - v_{DS} curve in the triode region. This process will continue until eventually the channel width becomes zero at the drain end. This pinch-off condition will occur when $v_{GD} \leq V_T$. To appreciate this fact, recall that the gate voltage had to be increased to V_T in order to create the channel. It follows that to make the channel disappear at the drain end the gate-to-drain voltage has to be reduced below V_T . In our case $v_{GS} = V_T + V$; thus pinch-off will happen when

$$v_{GD} \leq V_T \quad \text{or} \quad v_{DS} \geq V$$

Increasing the drain voltage above the value for pinch-off will not change the shape of the channel; hence the current i_D remains constant at the value reached at the onset of pinch-off.

Static Characteristics

The discussion above gives rise to the i_D - v_{DS} characteristics shown in Fig. 8.9, from which we see that there are two regions of operation: the triode region and the pinch-off region. The boundary between the two regions is determined from

$$v_{GD} = V_T \quad (8.1)$$

or, equivalently,

$$v_{DS} = v_{GS} - V_T \quad (8.2)$$

In the triode region the i_D - v_{DS} characteristics are described by the parabolic relationship

$$i_D = \beta[(v_{GS} - V_T)v_{DS} - \frac{1}{2}v_{DS}^2] \quad (8.3)$$

where

$$v_{GS} \geq V_T \quad \text{and} \quad v_{DS} \leq v_{GS} - V_T$$

The constant β depends, among other things, on the geometry of the device.

In the pinch-off region the current i_D is constant for a given value of v_{GS} , a fact that gives this region of operation the name *saturated region*. The current i_D is given by

$$i_D = \frac{1}{2}\beta(v_{GS} - V_T)^2 \quad (8.4)$$

where $v_{GS} \geq V_T$ and $v_{DS} \geq v_{GS} - V_T$. Equation (8.4) represents the graph shown in Fig. 8.10, from which the enhancement nature of the device should be evident. Specifically, at $v_{GS} = 0$ the current is almost zero, and for the device to conduct v_{GS} has to exceed the threshold voltage V_T .

Pinch-off occurs when

$$v_{DS} = v_{GS} - V_T$$

Substituting in Eq. (8.4) yields the equation of the boundary line (shown as a broken line in Fig. 8.9),

$$i_D = \frac{1}{2}\beta v_{DS}^2 \quad (8.5)$$

For modern MOS technology the threshold voltage V_T , which is a positive number for n -channel devices, lies in the range 1 to 3 V.

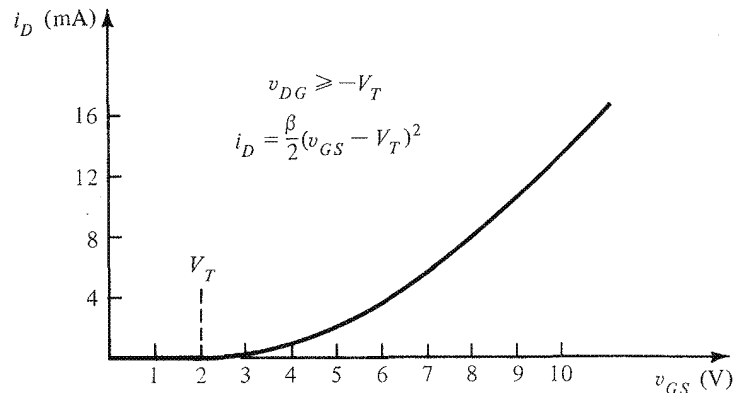


Fig. 8.10 The i_D - v_{GS} characteristic for an n -channel enhancement-type MOSFET in pinch-off ($V_T = 2$ V, $\beta = 0.5$ mA/V²).

Circuit Symbol

Figure 8.11 shows the circuit symbol of the *n*-channel enhancement-mode MOSFET. The only difference between this symbol and that of the depletion mode device is that the channel is represented by a broken line. This signifies that no channel exists in the enhancement device (until v_{GS} exceeds V_T).

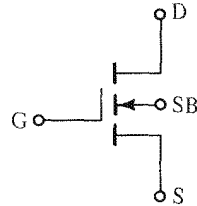


Fig. 8.11 Circuit symbol for the *n*-channel enhancement-type MOSFET.

The p-Channel Device

The operation of the *p*-channel MOSFET parallels that of the *n*-channel device described above. The characteristics are also similar except for a reversal of polarity of all currents and voltages. Also, V_T for the *p*-channel device is a negative number.

The characteristics of the *p*-channel MOSFET can be described by equations similar to Eqs. (8.1) through (8.5) except for replacing v_{GS} by v_{SG} , v_{DS} by v_{SD} , and V_T by $|V_T|$. The circuit symbol of the *p*-channel enhancement-mode MOSFET is shown in Fig. 8.12.

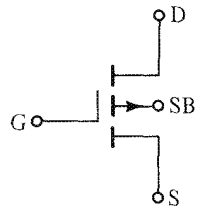


Fig. 8.12 Circuit symbol for the *p*-channel enhancement-type MOSFET.

The Role of the Substrate Bias

In most applications the substrate terminal is connected to the source terminal, which results in the *pn* junction between the substrate and the created channel being reverse-biased. In integrated circuits, however, the substrate is usually common to many MOS transistors. In order to maintain the reverse-bias condition the substrate should be connected to the most negative voltage (in the circuit) for an NMOS device and to the most positive voltage for a PMOS device. But what is the effect on device operation of the dc voltage between substrate and source? To answer this question, observe that the reverse-bias voltage between substrate and source can control the channel width in the same manner as the gate voltage in a JFET controls its operation. Thus increasing the reverse-bias voltage on the substrate causes the channel to be depleted of charge

carriers. This effect can be expressed as an increase in the magnitude of threshold voltage V_T as the substrate reverse bias is increased. An approximate rule for this is

$$\Delta V_T \approx C\sqrt{V_{SB}}$$

where V_{SB} is the source-to-substrate voltage and ΔV_T is the corresponding change in threshold voltage. The constant C depends on the doping of the substrate and generally lies in the range 0.5 to 2.0. Unless otherwise mentioned we shall assume that the substrate is connected to the source.

Temperature Effects

Both V_T and β are temperature sensitive. The magnitude of V_T decreases by about 2.5 mV for every rise of 1°C in temperature. This decrease in $|V_T|$ gives rise to a corresponding increase in drain current as temperature is increased. However, because β decreases with temperature and its effect is a dominant one, the overall observed effect of temperature increase is a *decrease* in drain current. This very interesting result is put to use in applying the MOSFET to power circuits.

Breakdown and Input Protection

In a MOSFET, breakdown occurs when the gate-to-source voltage exceeds about 100 V. However, because of the very high input impedance a small amount of static charge accumulating on the input capacitor (between gate and source) can cause this breakdown voltage to be exceeded. To make matters worse, breakdown of a MOSFET results in permanent damage to the oxide.

To prevent the accumulation of static charge on the input capacitor of a MOSFET, gate protection devices are usually included at the input of MOS integrated circuits (such as the input terminals of a CMOS logic gate). The protection mechanism invariably makes use of clamping diodes (see Fig. 6.50).

Simplified Circuit Symbols

To simplify matters, we shall adopt the simplified circuit symbols shown in Fig. 8.13. It is assumed in these symbols that the substrate is connected to the source. The arrow-

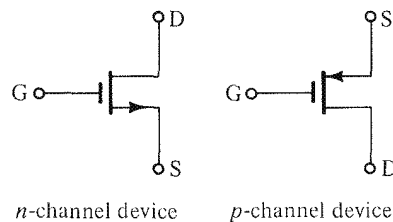


Fig. 8.13 Simplified circuit symbols for MOS transistors. The type of device (depletion or enhancement) will be made clear in each specific case.

head on the source points in the direction of normal current flow. Finally, note that the symbol for the p -channel device is drawn with the source up, which makes it conform with our circuit-drawing convention of currents flowing from top to bottom.

Recapitulation

Consider Fig. 8.14. For the n -channel enhancement device shown in the figure to conduct, v_{GS} has to be positive and greater than V_T . The device will operate in the pinch-off or active region if the drain voltage v_D is more positive than the gate voltage v_G by

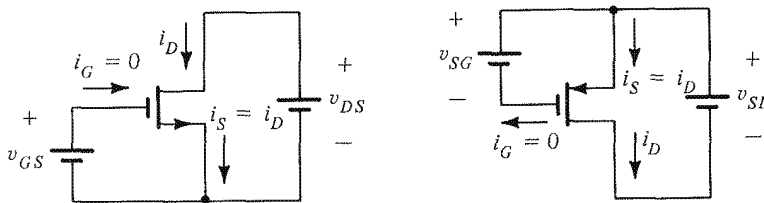


Fig. 8.14 Normal current flow directions and voltage polarities in enhancement MOS transistors.

at least $-V_T$. That is, the device will still be in pinch-off even if the drain voltage is lower than that of the gate by V_T volts. If the drain voltage is further reduced, the device gets out of pinch-off and goes into the triode region.

For the p -channel device to conduct, the source has to be made more positive than the gate by at least $|V_T|$ volts; that is, $v_{SG} \geq |V_T|$. The device will be in pinch-off as long as the drain voltage is lower than that of the gate, or even if it is higher than that of the gate by at most $|V_T|$. If v_D is increased more than $|V_T|$ volts above v_G , the device leaves pinch-off and enters the triode region.

Of course, the pinch-off region is the one suitable for amplifier application. Switching applications make use of the cutoff region and the triode region.

Finally, we should point out that real devices display a finite output resistance when operated in the pinch-off region. Figure 8.15 shows the i_D - v_{DS} characteristics of an n -channel MOS transistor of the enhancement type. In pinch-off the characteristic curves show finite slope that increases with the current level in the device.

EXERCISE

- 8.2** Consider an enhancement NMOS transistor with $V_T = 2$ V which conducts a current $i_D = 1$ mA when $v_{GS} = v_{DS} = 3$ V. What is the value of i_D for $v_{GS} = 4$ V and $v_{DS} = 5$ V? (Assume that in pinch-off the device acts as a current source.) Also calculate the value of the drain-to-source resistance r_{DS} for small v_{DS} and $v_{GS} = 4$ V.

Ans. 4 mA; 500 Ω

8.3 BIASING THE ENHANCEMENT MOSFET

As mentioned before, the first step in the design of a transistor amplifier involves establishing a stable and predictable dc operating point inside the active region of operation.